2D Gaussian Mixture Models, K-Means Clustering, and Classification

Lab Assignment 6 Report

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1 Objective

The main goals of this assignment are:

1. To Generate 2D synthetic data with 2 classes using Gaussian Mixture Models (GMMs) with $K = 3$ components per class.
2. And then to experiment with different covariance types across the components: isotropic ($\sigma^2\mathbf{I}$), diagonal, and full covariance matrices.
3. Then visualize the generated data points with different colors for each cluster.
4. Split the dataset into an 80% training set and a 20% test set for each class.
5. Finally perform K-Means clustering on the 80% training data to discover cluster centers and plot cluster boundaries.
6. Then classify the 20% test data using the learned cluster centers based on minimum distortion.
7. Evaluate the classification performance with confusion matrix and plot the resulting decision boundary.

2 Workflow

The experiment is executed in four straightforward steps:

1. **Data Generation:** Generated 4,000 2D points for Class 1 and 4,000 points for Class 2 (8,000 points in total). Each class consists of three Gaussian components with assigned weights, centers, and covariance shapes (isotropic, diagonal, and full), the parameters are empirically chosen. The points are then drawn using standard normal random numbers transformed by the mean and covariance of each component.
2. **Train-Test Split:** Each class is randomly shuffled and divided into:
   - **Training set (80%):** 3,200 points per class (used for clustering).
   - **Test set (20%):** 800 points per class (used for classification evaluation).
3. **K-Means Clustering on Training Data:** K-Means ($K = 3$) is run separately on the training data of Class 1 and Class 2. The algorithm automatically locates the three cluster centers for each class without any class or component labels. Here multiple random starts are used to ensure optimal convergence.
4. **Classification via Minimum Distortion:** For each test sample, the squared Euclidean distance is measured to the nearest cluster centeroid of Class 1 and to the nearest cluster centeroid of Class 2. After which the sample is then assigned to the class that gives the smallest distance (minimum distortion).

3 Outputs and Results

3.1 Parameters Used for Data Generation

Table 1 lists the chosen mixing weights, centers, and covariance matrices for the two classes.

Class & Component	Covariance Type	Weight	True Center (μ)	Covariance Matrix (C)
Class 1, Component 0	Isotropic	0.30	$[-9.0, 7.0]^T$	$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$
Class 1, Component 1	Diagonal	0.45	$[-9.0, 0.0]^T$	$\begin{bmatrix} 0.5 & 0.0 \\ 0.0 & 2.0 \end{bmatrix}$
Class 1, Component 2	Full	0.25	$[-9.0, -7.0]^T$	$\begin{bmatrix} 2.0 & 0.8 \\ 0.8 & 1.0 \end{bmatrix}$
Class 2, Component 0	Isotropic	0.35	$[-1.0, 7.0]^T$	$\begin{bmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix}$
Class 2, Component 1	Diagonal	0.25	$[-1.0, 0.0]^T$	$\begin{bmatrix} 1.5 & 0.0 \\ 0.0 & 0.7 \end{bmatrix}$
Class 2, Component 2	Full	0.40	$[-1.0, -7.0]^T$	$\begin{bmatrix} 1.5 & -0.6 \\ -0.6 & 1.0 \end{bmatrix}$

Table 1: Parameters for Class 1 and Class 2 components.

3.2 GMM Data Plot

Figure 1 displays the generated points for both the classes. Each component is plotted in a different color for distinction.

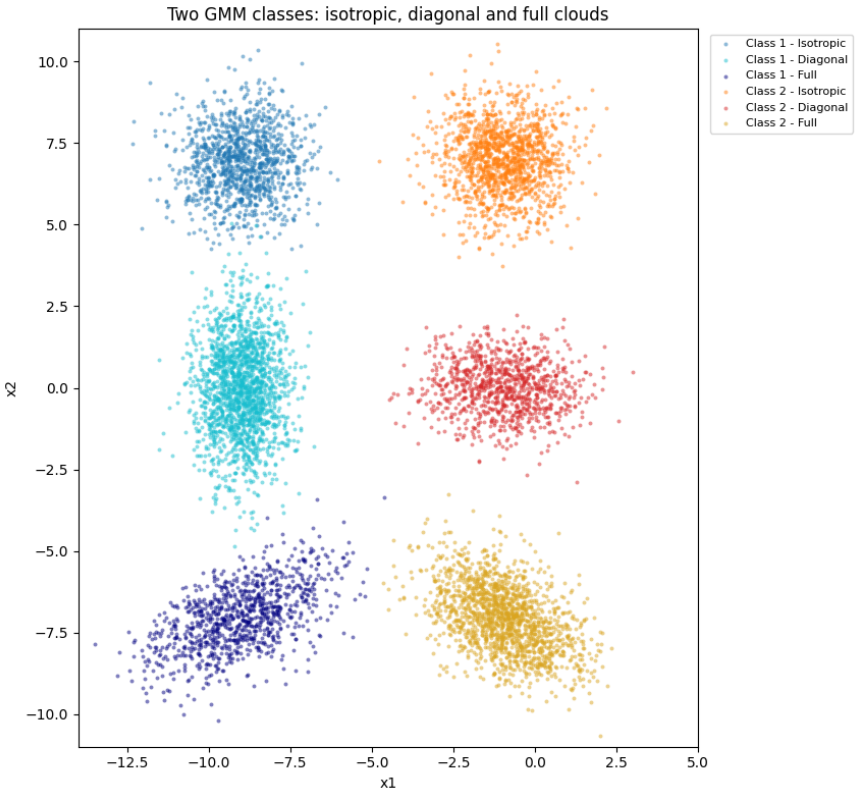


Figure 1: Generated 2D data for Class 1 (left) and Class 2 (right) across isotropic, diagonal, and full covariance components.

3.3 K-Means Clustering Outputs

Table 2 compares the original true means with the centers that are discovered by K-Means on the train data. Figure 2 shows the learned cluster boundaries and centeroids.

Class & Component	True Center	K-Means Discovered Center	Difference
Class 1, Top (Isotropic)	$[-9.00, \quad 7.00]^T$	$[-9.01, \quad 6.90]^T$	$[0.01, 0.10]^T$
Class 1, Middle (Diagonal)	$[-9.00, \quad 0.00]^T$	$[-9.00, -0.04]^T$	$[0.00, 0.04]^T$
Class 1, Bottom (Full)	$[-9.00, -7.00]^T$	$[-9.06, -7.02]^T$	$[0.06, 0.02]^T$
Class 2, Top (Isotropic)	$[-1.00, \quad 7.00]^T$	$[-0.97, \quad 7.00]^T$	$[0.03, 0.00]^T$
Class 2, Middle (Diagonal)	$[-1.00, \quad 0.00]^T$	$[-0.96, \quad 0.02]^T$	$[0.04, 0.02]^T$
Class 2, Bottom (Full)	$[-1.00, -7.00]^T$	$[-0.94, -7.07]^T$	$[0.06, 0.07]^T$

Table 2: True means versus learned K-Means cluster centers.

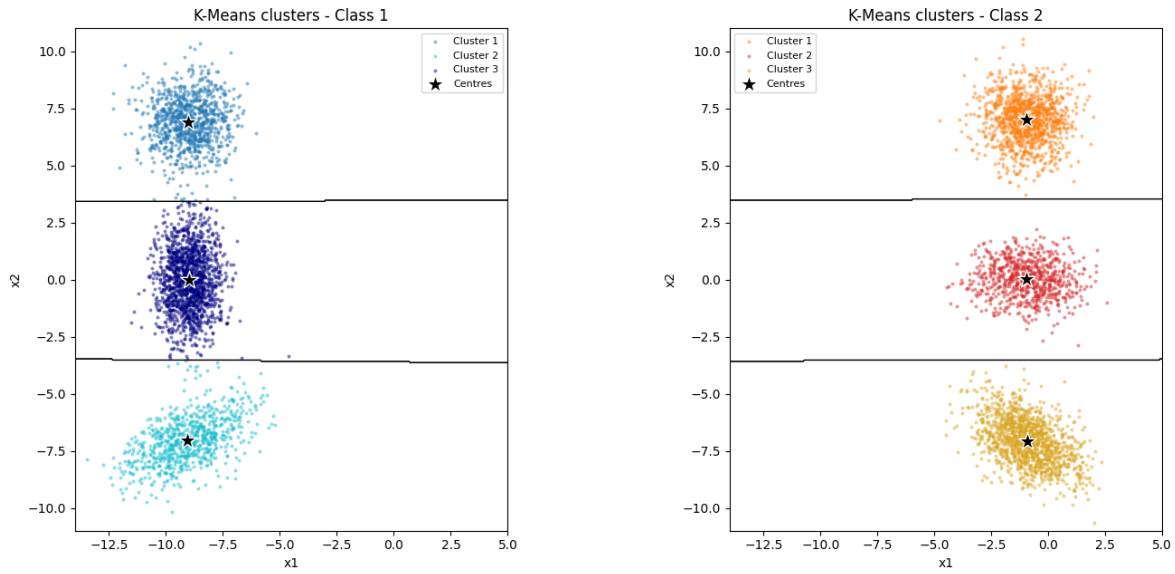


Figure 2: K-Means cluster assignments, centeroids (stars), and boundaries within each class (Class 1 (left) and Class 2 (right)).

3.4 Classification Results and Decision Boundary

Using the learned cluster centers, the 1,600 test samples were classified using minimum distortion. Table 3 presents the confusion matrix, and Figure 3 shows the resulting decision boundary.

True Class	Predicted Class 1	Predicted Class 2
Class 1	800	0
Class 2	0	800

Table 3: Confusion matrix on 1,600 test points (Test Accuracy: 100.0%).

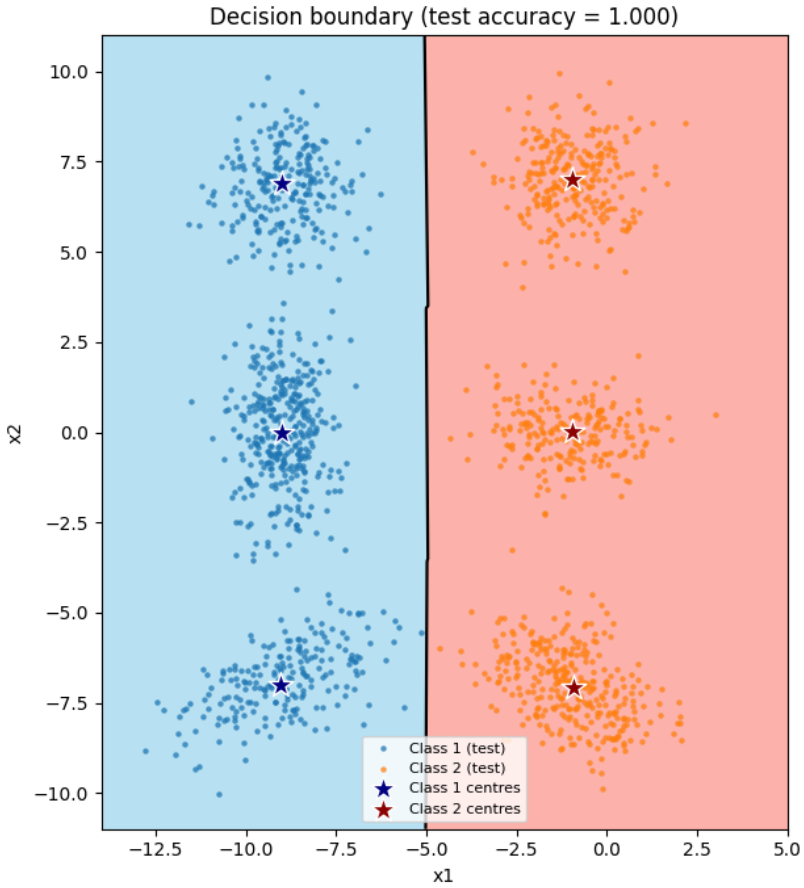


Figure 3: Decision regions (blue for Class 1, red for Class 2) and decision boundary at $x_1 \approx -5.0$ overlaid with test points and cluster centers.

4 Findings

- **Shape of Covariance Clouds:**
  - *Isotropic:* Equal spread in all directions produces circular clusters.
  - *Diagonal:* The unequal variances along the coordinate axes produce axis-aligned elliptical clusters (Class 1 is stretched vertically; Class 2 is stretched horizontally).
  - *Full Covariance:* The off-diagonal terms tilt the distribution diagonally (Class 1 tilts upward with positive correlation; Class 2 tilts downward with negative correlation).
- **Clustering performance of k-means:** K-Means successfully discovered all component centers without supervision. The discovered centers closely match the true parameters with an error of less than 0.1.
- **Behavior of decision boundary:** Because Class 1 centers are centered at $x_1 \approx -9.0$ and Class 2 centers are centered at $x_1 \approx -1.0$, the minimum distortion rule forms a vertical decision boundary at $x_1 \approx -5.0$ (midway between the two classes).
- **Classification Performance:** Both classes are cleanly separated, yielding zero test errors and a test accuracy of 100.0% ($1,600/1,600$ correct).